Exact Bézout Certificates and Coefficient Nonvanishing at Section Roots

Lean Stage 2 Algebraic Certificates

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Abstract

We record the Stage 2 Lean formalization of exact rational polynomial reduction and Bézout identities for nine explicit section-coefficient polynomials associated to a fixed degree-24 section modulus, together with the resulting nonvanishing theorems at every root of that modulus, the proof that the modulus has degree 24, the existence of a complex root, and the existence of a complex root at which all nine quantities are simultaneously nonzero. The module `Hodge.ExplicitResidueCertificates` contains 45 theorem declarations; combined with the seven Stage 1 lemmas this yields the audited total of 52 incremental theorems, each depending only on the standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. SymPy is used only to generate candidate witnesses; Lean proves every identity independently. No geometric cycle-class, Jacobian-quotient basis, or H/G resultant theorem is claimed.

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1 Scope and non-claims

This note is the companion to the Stage 1 determinant/derivative note. It covers every new Stage 2 theorem in `Hodge/ExplicitResidueCertificates.lean` at commit `e207394a9ed4255ad804860b8f69fde7e940dd54`, organized as one coherent algebraic certificate paper. Generated helper declarations (evaluation lemmas, reduction/Bézout pairs) are mapped as supporting infrastructure for the nine nonvanishing results and three nonvacuity theorems; they are not split into separate papers [1].

Proved here.

- Nine scalar nonvanishing theorems for the formulas w_0, w_1, w_2, w_3, u_0, u_1, u_2, u_3, and $\mathrm{disc} = b^2 - 4c$, under only the field / $\mathbb{Q}$-algebra structure and the modulus-root equation.
- Eighteen exact identities in $\mathbb{Q}[X]$: for each of the nine raw polynomials P, a reduction $P = R + Q \cdot m$ and a Bézout identity $AR + Bm = 1$.
- Degree 24 of the modulus polynomial, existence of a complex root, and simultaneous nonvanishing of all nine quantities at some complex root.

Not proved here.

- The tenth Python certificate: the actual H/G resultant and its common-projective-zero consequence. Pinned mathlib defines resultant via a Sylvester determinant and lists further resultant theory as TODO (`docs/FULL_FORMALIZATION_STAGE2.md`).
- Identification of the U formulas with coefficients in an actual Jacobian quotient (requires a Lean quotient-reduction/basis theorem).
- Complete-intersection ideal regularity, reducedness, saturation; geometric residue-cycle theorems; Betti/Hodge comparison; geometric group action; rational geometric spanning of Hodge classes.
- General or integral Hodge conjecture statements; novelty or peer-review claims.

2 Standing data

Definition 1 (Modulus and section coefficients)**.** Let $m \in \mathbb{Q}[X]$ be the explicit polynomial

$$m = X^{24} + 66081312000\,X^{18} - 4811512860000\,X^{12} + 1167566400000\,X^{6} + 583200000 \quad (1)$$

(`ExplicitResidueCertificates.modulus`). Let $b, c, d, e, g \in \mathbb{Q}[X]$ be the section polynomials `bPoly`,`cPoly`,`dPoly`,`ePoly`,`gPoly`, whose evaluations recover `ExplicitCurve.sb`,...,`sg`. For a characteristic-zero field K equipped with a $\mathbb{Q}$-algebra structure and $k \in K$, write $b(k) = \mathrm{sb}(k)$ etc., and define the nine scalar formulas

$$w_0 = 3bg - 2ce, \quad w_1 = 6g + be - 4cd, \quad w_2 = 4e - bd - 6c, \quad w_3 = 2d - 3b, \tag{2}$$

$$u_0 = cw_0 - 6, \quad u_1 = bw_0 + cw_1, \quad u_2 = w_0 + bw_1 + cw_2, \quad u_3 = w_1 + bw_2 + ck, \tag{3}$$

$$\mathrm{disc} = b^2 - 4c. \tag{4}$$

The corresponding elements of $\mathbb{Q}[X]$ are `w0Poly`,...,`discPoly`.

Remark 2 (Trust boundary for SymPy)**.** The script `scripts/generate_residue_certificates.py` uses SymPy 1.14.0 [4] to compute candidate sparse rational polynomials R, Q, A, B. Those candidates are written into Lean source; every equality is then proved by Lean polynomial extensionality and exact ring normalization. SymPy is reproducible generation tooling, not part of the proof trust base (`docs/FULL_FORMALIZATION_STAGE2.md`).

3 Supporting polynomial identities

Lemma 3 (Evaluation)**.** *For each section polynomial $P \in \{b, c, d, e, g\}$ and each raw coefficient polynomial $P \in \{w_0, \ldots, w_3, u_0, \ldots, u_3, \mathrm{disc}\}$,*

$$\mathrm{aeval}_k(P_{\mathrm{Poly}}) = P(k) \tag{5}$$

in every characteristic-zero field K equipped with a $\mathbb{Q}$-algebra structure. Lean declares the hypotheses `[Field K]`, `[CharZero K]`, and `[Algebra ℚ K]`. In particular the five section polynomials evaluate to `ExplicitCurve.sb`,...,`sg`.

Lean. Fourteen `@[simp]` theorems (`eval_bPoly` through `eval_disc`) are proved under those field and $\mathbb{Q}$-algebra hypotheses by unfolding definitions and applying the algebra-map simp set. □

Lemma 4 (Reduction and Bézout)**.** *For each of the nine raw polynomials $P \in \mathbb{Q}[X]$ there exist explicitly defined sparse $R, Q, A, B \in \mathbb{Q}[X]$ such that*

$$P = R + Q \cdot m \qquad \text{and} \qquad AR + Bm = 1 \tag{6}$$

as equalities in $\mathbb{Q}[X]$.

Lean. Eighteen theorems `w0_reduction`, `w0_bezout`, ..., `disc_reduction`, `disc_bezout` are proved by `Polynomial.funext`, evaluation at a generic point, and `norm_num`/`ring`. Exact rational coefficients are recorded in the Lean source; they are omitted here only for readability. □

Lemma 5 (Evaluation criterion)**.** *If $P = R + Qm$ and $AR + Bm = 1$ in $\mathbb{Q}[X]$, and if k is a root of m in a characteristic-zero field K equipped with a $\mathbb{Q}$-algebra structure (Lean: `[Field K]`, `[CharZero K]`, `[Algebra ℚ K]`), then* $\mathrm{aeval}_k(P) \neq 0$.

Lean. `eval_ne_zero_of_certificate`, under the same field/$\mathbb{Q}$-algebra hypotheses: evaluate both identities at k; the Bézout identity collapses to $A(k)R(k) = 1$, so $R(k) \neq 0$, hence $P(k) \neq 0$. □

4 Main nonvanishing theorems

Theorem 6 (Nine coefficient nonvanishings)**.** *Let K be a field of characteristic zero with a $\mathbb{Q}$-algebra structure, and let $k \in K$ satisfy*

$$k^{24} + 66081312000\,k^{18} - 4811512860000\,k^{12} + 1167566400000\,k^{6} + 583200000 = 0. \tag{7}$$

Then

$$w_0(k),\ w_1(k),\ w_2(k),\ w_3(k),\ u_0(k),\ u_1(k),\ u_2(k),\ u_3(k),\ \mathrm{disc}(k) \tag{8}$$

are all nonzero.

Lean. Nine theorems `w0_ne_zero`,…,`disc_ne_zero`. Each rewrites the scalar hypothesis as $\mathrm{aeval}_k(m) = 0$ and applies Lemma 5 to the matching reduction/Bézout pair. No numerical root approximation, irreducibility hypothesis, geometric datum, or desired nonvanishing is assumed. □

Theorem 7 (Modulus degree)**.** $\deg(m) = 24$.

Lean. `modulus_degree`: unfold m, apply `compute_degree`, and discharge closed numerical bounds by `norm_num` and ordinary `decide +kernel`. This is not a `native_decide` shortcut on the algebraic content of the nonvanishing theorems. □

Theorem 8 (Complex root existence)**.** *There exists $k \in \mathbb{C}$ with* $\mathrm{aeval}_k(m) = 0$.

Lean. `exists_complex_root`: apply algebraic closedness of $\mathbb{C}$ (`IsAlgClosed.exists_aeval_eq_zero`) using Theorem 7 and $\deg m \neq 0$. □

Theorem 9 (Simultaneous nonvanishing at a complex root)**.** *There exists $k \in \mathbb{C}$ satisfying the modulus equation such that*

$$w_0(k) \neq 0,\ \ldots,\ w_3(k) \neq 0,\ u_0(k) \neq 0,\ \ldots,\ u_3(k) \neq 0,\ \mathrm{disc}(k) \neq 0. \tag{9}$$

Lean. `exists_complex_nonvanishing`: take a root from Theorem 8 and apply all nine cases of Theorem 6. This shows the universal nonvanishing statements are not vacuous assertions over a field without section roots. It is not a geometric cycle-class assertion. □

5 Theorem–declaration mapping

Main results

#	Lean declaration	Role
1	`w0_ne_zero`	$w_0 = 3bg - 2ce \neq 0$
2	`w1_ne_zero`	$w_1 = 6g + be - 4cd \neq 0$
3	`w2_ne_zero`	$w_2 = 4e - bd - 6c \neq 0$
4	`w3_ne_zero`	$w_3 = 2d - 3b \neq 0$
5	`u0_ne_zero`	$u_0 = cw_0 - 6 \neq 0$
6	`u1_ne_zero`	$u_1 = bw_0 + cw_1 \neq 0$
7	`u2_ne_zero`	$u_2 = w_0 + bw_1 + cw_2 \neq 0$
8	`u3_ne_zero`	$u_3 = w_1 + bw_2 + ck \neq 0$

9	`disc_ne_zero`	$b^2 - 4c \neq 0$
10	`modulus_degree`	$\deg m = 24$
11	`exists_complex_root`	$\exists\, k \in \mathbb{C},\ m(k) = 0$
12	`exists_complex_nonvanishing`	simultaneous nonvanishing

Supporting certificates (same paper; not split)

Count	Lean declarations	Role
5	`eval_bPoly...eval_gPoly`	section evaluation
9	`eval_w0...eval_disc`	raw-coefficient evaluation
9	`*_reduction`	$P = R + Qm$ in $\mathbb{Q}[X]$
9	`*_bezout`	$AR + Bm = 1$ in $\mathbb{Q}[X]$
1	`eval_ne_zero_of_certificate`	evaluation criterion

Total Stage 2 theorems: $12+5+9+9+9+1 = 45$. Combined with the seven Stage 1 lemmas: 52 audited declarations (`ResidueAxiomAudit.lean`).

6 Axiom audit and verification evidence

The checker `scripts/check_residue_axioms.py` requires `ResidueAxiomAudit.lean` to enumerate every theorem in both incremental modules in source order, builds `Hodge.ExplicitResidueAlgebra` and `Hodge.ExplicitResidueCertificates`, and accepts only `propext`, `Classical.choice`, `Quot.sound`.

Verified log lines (America/Chicago evidence timestamps on disk):

- `build-and-audit-final.log`: reports successful `Hodge.ExplicitResidueCertificates` build (7085/7085) and "PASS: 52 incremental theorem declarations; standard axioms only".
- `residue-audit-e207394.log`: same PASS line; 52 axiom-print entries.
- `no-sorry-e207394.log`: "OK: no sorry/admit/native_decide, no axiom declarations".
- `full-build-e207394.log`: full repository `lake build` completed successfully at commit `e207394`.
- Prior regeneration check (certificate delivery evidence): byte-identical SymPy regeneration of `ExplicitResidueCertificates.lean`; `LICENSE`, `CITATION.cff`, and Stage 1 Lean source unchanged.

7 Toolchain pins and reproducibility

Pin	Value
Lean [2]	`leanprover/lean4:v4.22.0`
mathlib input [3]	`v4.22.0`
mathlib checkout	`79e94a093aff4a60fb1b1f92d9681e407124c2ca`
SymPy (generator only) [4]	1.14.0
Git commit	`e207394a9ed4255ad804860b8f69fde7e940dd54`
PR	https://github.com/BenFrohman/HODGE/pull/36
License	Apache-2.0 (preserved; no relicense)

```
python3 scripts/generate_residue_certificates.py
# optional: git diff --exit-code -- Hodge/ExplicitResidueCertificates.lean
lake build Hodge.ExplicitResidueAlgebra \
           Hodge.ExplicitResidueCertificates
python3 scripts/check_residue_axioms.py
python3 scripts/check_no_sorry.py
```

Local sources and evidence inspected (read-only; no repository edits). Mathematical source pin `e207394a9ed4255ad804860b8f69fde7e940dd54`; later inspected worktree `fefe2ba9d87c900b538d02c13f7f51f375fcd285` retains identical reviewed modules, audit, and toolchain/lock files:

- `/workspace/hodge-formal-stage1-20261106/Hodge/ExplicitResidueCertificates.lean`
- `/workspace/hodge-formal-stage1-20261106/ResidueAxiomAudit.lean`
- `/workspace/hodge-formal-stage1-20261106/docs/FULL_FORMALIZATION_STAGE2.md`
- `/workspace/hodge-formal-stage1-20261106/scripts/generate_residue_certificates.py`
- `/workspace/hodge-formal-stage2-evidence-20261106/` (actual files: `build-and-audit-final.log`, `build-and-audit.log`, `full-build.log`, `typecheck*.log`, `DegreeCheck.lean`, `generate_certificates.py`)
- `/workspace/hodge-formal-stage1-evidence-20261106/` (`Stage1Audit.lean`, `axioms.txt`, `baseline-explicit-curve.lean`)
- `/workspace/hodge-certificate-delivery-20261106/` (`full-build-e207394.log`, `residue-audit-e207394.log`, `no-sorry-e207394.log`, status JSON).

CI verification at a later identical head. User-reported merge of PR https://github.com/BenFrohman/HODGE/pull/36 is acknowledged; this packaging records no merge-commit SHA (merge verification is underway separately). At head `fefe2ba9d87c900b538d02c13f7f51f375fcd285`, where the reviewed Lean modules, audit, and toolchain/lock match mathematical pin `e207394a9ed4255ad804860b8f69fde7e940dd54`, all five reported GitHub Actions checks completed successfully: Lean/build and Lean/attempt (https://github.com/BenFrohman/HODGE/actions/runs/42317080169),

Lean certificate/certify on the PR (https://github.com/BenFrohman/HODGE/actions/runs/42317080616), Lean certificate/certify on the push (https://github.com/BenFrohman/HODGE/actions/runs/42317075046), and Paper/paper (https://github.com/BenFrohman/HODGE/actions/runs/42317080271). Both certification logs build the incremental modules and print `PASS: 52 incremental theorem declarations; standard axioms only`. CI success verifies those algebraic declarations; it does not discharge geometric Hodge/Betti/cycle-class obligations. This note remains a REVIEW DRAFT with Zenodo draft DOI `10.5281/zenodo.24025150` reserved; unpublished (not a live citation / not a publication claim).

8 Limitations relative to the geometric program

The nine nonvanishings are algebraic statements about evaluations of explicit elements of $\mathbb{Q}[X]$ at roots of m. They do not by themselves produce geometric residues, nor do they prove that the corresponding Jacobian-quotient classes are nonzero in a geometric cohomology theory. Villaflor Loyola's complete-intersection residue theorem [5] and Griffiths' comparison remain external classical input (`NOTICE.md`).

The classical Python projection certificate checks ten exact unit-at-every-root identities (eight W/U formulas, the discriminant, and the H/G resultant). Lean Stage 2 covers the first nine; the tenth (resultant) remains unformalized.

These algebraic certificates can stand as fully formalized Lean mathematics while a full geometric Hodge paper for the sextic fourfold remains unpublished pending the missing geometric foundations. Zenodo draft DOI reserved: `10.5281/zenodo.24025150` (https://doi.org/10.5281/zenodo.24025150); **unpublished** — not a live citation and not a publication.

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References

[1] Benjamin Stanley Frohman. HODGE: Lean 4 formalization and hodge ledger. https://github.com/BenFrohman/HODGE, 2026. Source commit `e207394a9ed4255ad804860b8f69fde7e940dd54`; PR https://github.com/BenFrohman/HODGE/pull/36. Historical archive DOI `10.5281/zenodo.23629480` identifies prior repository provenance and is not a DOI for these incremental algebra notes.

[2] Lean FRO. The Lean 4 theorem prover. https://github.com/leanprover/lean4, 2026. Pinned release `v4.22.0` via `lean-toolchain`.

[3] mathlib community. mathlib4. https://github.com/leanprover-community/mathlib4, 2026. Pinned input revision `v4.22.0`; locked checkout `79e94a093aff4a60fb1b1f92d9681e407124c2ca` from `lake-manifest.json`.

[4] Aaron Meurer et al. SymPy: symbolic computing in Python, 2026. Used only to generate candidate rational witnesses; not part of the Lean proof trust base.

[5] Roberto Villaflor Loyola. Periods of complete intersection algebraic cycles. *Manuscripta Mathematica*, 167:765–792, 2022. Cited as classical residue-cycle input; not formalized in the incremental Lean modules of this note.